Supplement to: Channel Estimation for Rydberg Receivers Using Learned Hankel Filtering

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***Abstract*—This supplement explains what the measurements can determine, the cost and limits of the filter, and the tests added in response to review. The original trained models and main-study results are unchanged. New small tests are exploratory: they do not establish a new robustness guarantee or replace the original confidence intervals.**

I. Local Recovery and Reference Calibration

For one array element, let $g, s_p \in \mathbb{C}^K$, with $\mu_p = g^T s_p + b_p$ and noiseless amplitude $a_p = |\mu_p|$. Use real coordinates $x = [\operatorname{Re} g^T, \operatorname{Im} g^T]^T$. When every $\mu_p \neq 0$, differentiation gives

$$da_p = \operatorname{Re}\{\mu_p^* s_p^T dg/|\mu_p|\}, \tag{1}$$

$$A_p = [\operatorname{Re} c_p^T, -\operatorname{Im} c_p^T], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{2}$$

Here $A \in \mathbb{R}^{P\times 2K}$ is the real amplitude Jacobian.

Local statement. If $\operatorname{rank} A = 2K$, the noiseless amplitude map is one-to-one in a sufficiently small neighborhood of this g. To see this, choose $2K$ independent rows. Their square Jacobian is invertible, so the inverse-function theorem supplies a local inverse for that measurement subset. The remaining measurements cannot remove this local injectivity. This is not a global uniqueness theorem: distant channels may give the same measurements. A rank-deficient Jacobian also does not, by itself, rule out all forms of nonlinear local injectivity.

For small perturbations at a full-rank point, least-squares inversion has first-order sensitivity bounded by $1/\sigma_{\min}(A)$. Thus $P \geq 2K$ is necessary for regular full-rank local recovery of an unrestricted row. It is not sufficient. Full row rank of S permits the EM linear solve but does not ensure full rank of A. These are standard local arguments in an affine phase-retrieval setting, not a new identifiability theorem [1]. Sparse channel constraints require their own parameterization and Jacobian; a path-count condition is not derived here.

A. Rician Information

For known complex variance $v > 0$ and $a > 0$, the amplitude score is

$$h(z, a, v) = -2a/v + (2z/v)I_1(2za/v)/I_0(2za/v). \tag{3}$$

Define $w(a, v) = \mathbb{E}[h(Z, a, v)^2]$, with expectation under the Rician observation law. For independent pilots the real Fisher matrix is

$$J = A^T \operatorname{diag}\{w(a_p, v)\}_{p=1}^P A. \tag{4}$$

Each magnitude observation contributes an outer product of rank at most one, not a full-rank complex information block. For finite $v > 0$ and nonzero amplitudes these weights are positive. At a zero field the regular amplitude-coordinate derivation is not used; first-order information about the Cartesian mean vanishes. When the assumptions hold, the rank and conditioning of J give a local diagnostic. Neither a small bound nor a well-conditioned matrix at the true channel guarantees that an iterative estimator reaches the correct solution.

B. What a Reference Does and Does Not Fix

With $B = 0$, g and $e^{j\phi}g$ have identical amplitudes for every ϕ. The Jacobian has the phase direction jg in its null space for nonzero g. A nonzero known reference can remove that symmetry, but arbitrary reference and pilot patterns still require a rank check. For example, if $b_p = c^T s_p$ for some fixed c, then $g' = e^{j\phi}(g+c) - c$ produces the same amplitudes. This can occur with a full-row-rank pilot matrix and a nonzero reference. RSR 0 dB in our experiments means a nonzero reference equal in expected power to one user; it is not the absent-reference case.

There is a separate calibration ambiguity. If the estimator assumes $B' = e^{j\phi}B$, then $G' = e^{j\phi}G$ gives

$$|G'S + B'| = |e^{j\phi}(GS + B)| = |GS + B|. \tag{5}$$

The likelihood therefore cannot identify a constant reference-phase error from these observations alone. With error measured in the original reference frame, even this otherwise exact rotated estimate has NMSE $|e^{j\phi} - 1|^2 = 4\sin^2(\phi/2)$. At 10° this is about 0.0304 (-15.2 dB). This is an illustrative exact-fit ambiguity, not a lower bound on every biased estimator. A smaller withheld loss cannot certify correct absolute phase or recover an unknown reference phase. Time-varying phase errors generally cannot be absorbed into one constant channel rotation. Joint reference calibration would need additional assumptions or measurements; it is not part of the proposed estimator.

II. Filter Properties and Computation

Let $H = U \operatorname{diag}(\sigma_i) V^H \in \mathbb{C}^{m\times q}$, $m \geq q$, and

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \qquad \tau > 0. \tag{6}$$

Its singular values are $f_\tau(\sigma) = \sigma^3/(\sigma^2 + \tau^2)$. Thus $0 \leq f_\tau(\sigma) \leq \sigma$ and

$$\|F_\tau(H)\|_F \leq \|H\|_F, \qquad \|F_\tau(H) - H\|_F \leq \sqrt{q}\,\tau/2. \tag{7}$$

The second inequality follows because $\sigma - f_\tau(\sigma) = \sigma\tau^2/(\sigma^2 + \tau^2) \leq \tau/2$.

For a channel vector, let d_i count its occurrences in the Hankel matrix and define $\|g\|_D^2 = \sum_i d_i |g_i|^2 = \|\mathcal{H}(g)\|_F^2$. Antidiagonal averaging followed by lifting is the orthogonal

projection onto Hankel matrices in Frobenius norm. Therefore for fixed realized $\eta \in [0,1]$ and $\tau > 0$,

$$g^+ = (1-\eta)g + \eta \mathcal{H}^\dagger F_\tau(\mathcal{H}(g)), \tag{8}$$

$$\|g^+\|_D \le \|g\|_D, \qquad \|g^+ - g\|_D \le \eta\sqrt{q}\,\tau/2. \tag{9}$$

This bounds the size of one correction, not its error relative to the unknown true channel. It does not prove a contraction between two different inputs, particularly when the controller changes η and τ with the input. It also does not prove convergence of the full estimator. For fixed H, $\tau \to 0$ gives $F_\tau(H) \to H$ and $\tau \to \infty$ gives zero. No low-noise limit is asserted because the learned threshold need not vanish with noise.

The matrix-solve shrinker is smooth for $\tau > 0$, including repeated singular values. However, the controller also uses ordered largest and tail singular values. These features need not be differentiable at every singular-value tie. Finite-gradient numerical checks do not establish global differentiability of the whole network.

A. Relaxed EM Is Not the Same as the Full Stage

For fixed posterior field Q and supplied variance $v > 0$, the negative EM surrogate, up to constants, is $\Psi(G) = \|GS + B - Q\|_F^2/v$. Its minimizer is G_{EM} when S has full row rank. The normal equations give

$$\Psi(G) - \Psi(G_{\mathrm{EM}}) = \|(G - G_{\mathrm{EM}})S\|_F^2/v. \tag{10}$$

For $G_\gamma = G + \gamma(G_{\mathrm{EM}} - G)$, the surrogate decrease is

$$\Psi(G) - \Psi(G_\gamma) = \gamma(2-\gamma)\|(G - G_{\mathrm{EM}})S\|_F^2/v \ge 0 \tag{11}$$

for $0 \le \gamma \le 2$. Thus even an over-relaxed step within this interval is a generalized EM step for this quadratic M-step, in exact arithmetic. The supplied-model likelihood is nondecreasing. This result does not extend automatically to the subsequent Hankel correction, nor does it guarantee channel-error reduction, global recovery or convergence of the complete learned estimator. The secondary GD core is not covered by this particular argument.

B. Operation Counts

For $S \in \mathbb{C}^{K\times P}$ and $G \in \mathbb{C}^{N\times K}$, forming SS^H costs $O(K^2P)$. Spectral initialization forms N weighted $(K+1)$-dimensional Gram matrices and eigendecompositions, costing $O(NP(K+1)^2 + N(K+1)^3)$. These are per observation case, not per stage. The inverse-trace noise proxy adds $O(K^3)$ once.

Here $q = \lceil N/2 \rceil$, $m = N - q + 1$. Caching the pilot-Gram factorization could remove repeated K^3 factorizations, but the quoted timings use the existing implementation. The filter avoids differentiating hard rank truncation; it does not remove the SVD needed for controller features. At large N, both dense spectral features and the dense solve scale cubically. A small parameter count is not a small operation count.

For HS-GS, with $R = q$ candidate ranks, $T_s = 25$ selection updates, $T_f = 100$ refit updates and $C = 4$ sweeps, the dominant repeated work is of order

$$(RT_s + T_f)\,[NKP + NK^2 + K^3 + CKmq^2]. \tag{12}$$

TABLE I
LEADING WORK PER STAGE, EXCLUDING INITIALIZATION

Operation	Leading order
Predicted fields and EM products	$O(NKP)$
EM solve, current implementation	$O(NK^2 + K^3)$
Spectral controller features	$O(Kmq^2)$
Smooth matrix filter	$O(Kmq^2 + Kq^3)$
Antidiagonal averaging	$O(Kmq)$
Small controller	$O(K \cdot 178)$

TABLE II
ORIGINAL TRANSFER RESULTS: GAIN OVER HS-GS IN DB

Changed quantity	Gain	Pointwise 95% interval
$N = 8$	-0.106	$[-0.286, 0.077]$
$N = 16$	1.223	$[1.086, 1.361]$
$N = 64$	0.165	$[0.024, 0.307]$
$K = 2$	0.078	$[-0.073, 0.228]$
$K = 4$	1.529	$[1.415, 1.639]$
$P = 10$	2.334	$[2.136, 2.531]$
$P = 20$	1.427	$[1.273, 1.577]$
Noise input -3 dB	1.226	$[1.095, 1.359]$
Noise input $+3$ dB	1.050	$[0.901, 1.193]$

Selection uses fewer pilot columns and has additional scoring and initialization work. This expression is a work estimate, not an exact FLOP count or a prediction of the measured speedup. Bessel functions, library choices, memory traffic and batching also matter. At $N = 32$, HS-GS uses 400 selection and 100 refit updates, compared with ten learned stages. Accuracy at those different budgets, not equal iteration count, is the reported comparison.

III. Existing Dimension Tests and Controller Inputs

The original study already contains one-factor-at-a-time tests, with 400 cases per condition and weights fixed. Except for the changed quantity, the reference condition is $N = 32, K = 3, P = 30$, SNR 5 dB and RSR 12 dB. The pilot sweep's smallest value is ten for three users, not six. It does not test the boundary $P = 2K$.

The full controller receives a spectrum summary, residual/noise proxies and size/stage information. A large spectral tail can reflect either noise or real paths; residuals can reflect noise, a wrong channel, or a wrong receiver model. The features provide context but no unambiguous failure detector. No sign constraint makes the threshold monotone in SNR, RSR or path count. The threshold and mixing can compensate for each other, so we inspect the actual relative channel correction as well as both outputs.

A. Original Follow-up Comparisons

The earlier follow-up trains 15 models and evaluates two fresh sets, each with seven SNR settings and two difficult settings of 200 cases. The six primary comparisons adjust for

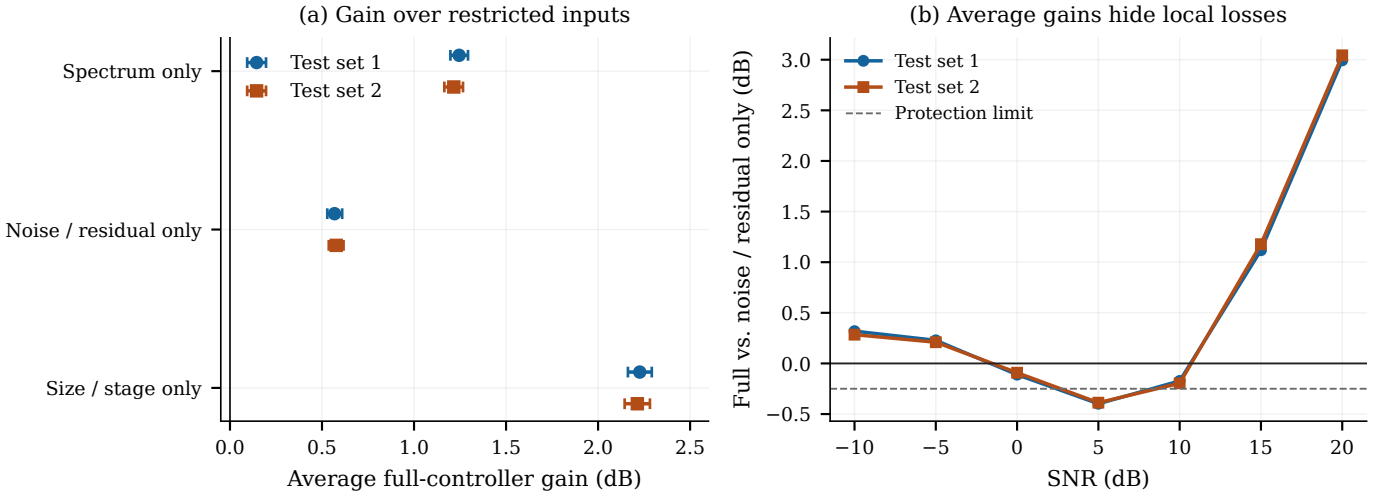


Fig. 1. Original smaller-budget input ablation, unchanged. Full inputs help the average but fail the declared 0.25 dB protection condition. These are separate models trained with 4,096 cases, 12 epochs and three seeds; they are not replacements for the main 16,384-case models.

TABLE III
ORIGINAL FOLLOW-UP AVERAGE GAINS IN DB, UNCHANGED

Comparison	Set 1	Set 2
New-data adaptive vs. global	2.252	2.185
New-data adaptive vs. no filtering	4.250	4.191
New-data adaptive vs. HS-GS	1.939	1.837
Original adaptive vs. supervised	1.015	1.012
Original adaptive vs. final-only	0.697	0.694
Fixed-open mix vs. original adaptive	0.064	0.066

multiple testing within each set. The two new-data adaptive runs use independently generated training data, one fixed training seed per dataset. They are distinct from the three main seeds sharing one dataset and from the new reviewer diagnostics.

The supervised recipe holds architecture, input pilots and training budget fixed but trains on channel targets with mean linear NMSE. Selecting its checkpoint by withheld likelihood reduces the gap to approximately 0.73 dB; pooling linear errors across SNRs reduces it to about 0.36 dB. It wins in some cases, including two paths. Its transfer from 21 to 30 pilots was less consistent in validation replay. This comparison does not establish a general benefit of withholding labels. The final-only comparison retrains that alternative and its separate timing sample gives 4.65 versus 2.05 ms; those values do not replace the main timing table. Fixing mixing at one retains all 188 stored parameters and worsens two of three seeds on two-path cases, despite its small average gain.

IV. Additional Reviewer Diagnostics

These checks were specified after the main study and use frozen selected EM models (seeds 9101–9103). No learned weights or original results were changed. Seven SNR cells and two stress cells each contain 32 fresh worlds: 288 distinct worlds. Eight input-mismatch conditions reuse the same 32 SNR 5 dB worlds, giving 544 condition-cases, not 544 independent worlds. Classical and learned methods share the channel, pilots and noise. All tests use $N = 32, K = 3, P = 30$; the main RSR is 12 dB. The extra cells have RSR 0 dB or two paths, at SNR 5 dB.

The small sample, post-review design and fixed checkpoints make these exploratory checks. The intervals resample whole cases with common weights across methods and learned seeds, separately within each cell. They are unadjusted, conditional intervals, not a new confirmatory success test. The original 400-case cells and their intervals remain unchanged.

A. Same Candidates, Different Rank-Selection Loss

HS-GS fits ranks 1–16 on the first 21 pilots using 25 EM updates and four Cadzow sweeps. Exactly the same candidate channels are scored on the last nine pilots using either squared magnitude residuals or Rician negative log-likelihood. Each selected rank is refitted with 100 updates and all 30 pilots. This changes only the selection score, not the fit budget or observation count. A replay against the original residual-ranked solver agreed exactly on a fresh case; an independent likelihood check agrees with SciPy's Rician density up to observation-only constants.

Across the seven SNR cells, Rician scoring improves HS-GS by 0.182 dB, with interval $[0.104, 0.267]$ dB. The frozen learned estimator retains a gain of 1.282 dB, with interval $[1.092, 1.485]$ dB. This score difference explains part, but not all, of the gap in this small check.

B. A Stronger Final-Filter Control

A separate bank of 72 development worlds (eight per SNR/stress cell) selects one global setting for each final-filter method. Selection uses only the withheld Rician likelihood after EM100 on 21 pilots. Hard truncation and OptShrink search ranks 1–15; cubic filtering searches relative thresholds $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$. Every method also has a no-correction option. Candidate counts differ and the grids are small; this is not an exhaustive equal-search-budget comparison. The selected settings are rank six, relative threshold 0.3

TABLE IV
EXPLORATORY SEVEN-SNR AVERAGES, 32 FRESH CASES PER CELL

Method	NMSE (dB)
EM100, no final filter	-10.162
EM100 + final hard rank six	-11.811
EM100 + final cubic filter	-11.648
EM100 + final OptShrink rank seven	-13.691
HS-GS, residual rank score	-12.778
HS-GS, Rician rank score	-12.961
Frozen learned model, stage-five prefix	-13.353
Frozen learned model, all ten stages	-14.243

and OptShrink rank seven, respectively. The settings are frozen before reading test observations, then used after EM100 with all 30 pilots. None use true channels for tuning.

OptShrink uses the empirical spectral-tail transform in the original algorithm [2], including rectangular zero directions and reduced tail dimensions. A rank equal to full capacity has no tail and is excluded; no correction is a separate candidate. At a tied boundary within a relative squared gap of 10^{-13}, the affected weight is set to its zero-weight limit. Primitive checks cover zero tail, scaling, transpose symmetry, direct rectangular trace evaluation, zero input and the tie rule. These are numerical safeguards, not a new optimality result.

In the matched one-pass test, OptShrink beats the cubic filter by 2.042 dB on the seven-SNR average. Cubic shrinkage is therefore not generally the best structural operation. The complete frozen learned estimator has a 0.552 dB average gain over EM100 plus OptShrink, with conditional interval $[0.432, 0.673]$ dB. It loses 1.040 dB at SNR 20 dB and 2.408 dB at RSR 0 dB, but gains 1.830 dB on two-path cases against this globally selected rank-seven control. That last result does not remove the two-path loss against per-case rank-selected HS-GS.

These comparisons have different update budgets, training/development budgets and filtering schedules. They do not identify the best possible OptShrink-based estimator or reproduce iterative LRHD. No controlled runtime comparison was added; the original 64-fold timing ratio applies only to the original HS-GS implementation.

C. Reference and Noise Inputs

The physical observations stay fixed. For amplitude errors the estimator receives $B_{\text{in}} = aB_{\text{true}}$ with $a = 0.9$ or 1.1; these are amplitude, not power, multipliers. Constant phase errors use $B_{\text{in}} = e^{j\delta}B_{\text{true}}$ with $\delta = \pm 10^\circ$. The ramp uses $B_{\text{in},np} = B_{\text{true},np}e^{j\delta p/(P-1)}$, $p = 0, \ldots, P-1$: the assumed phase error grows from zero to $\pm 10^\circ$. Noise errors supply $v_{\text{in}} = v_{\text{true}}10^{\Delta/10}$, $\Delta = \pm 6$ dB. These are input-mismatch stress tests, not measured hardware drift. They test errors relative to a fixed true reference, not a dynamically changing atomic receiver or a joint calibration algorithm.

Supplying noise variance 6 dB too high worsens the learned estimate by 3.048 dB relative to matched inputs, and makes it 1.409 dB worse than Rician-ranked HS-GS on these paired cases. This is a new observed failure, not evidence of general variance robustness. Amplitude errors also cause clear losses.

TABLE V
INPUT MISMATCH AT SNR 5 DB: LOWER NMSE IS BETTER

Assumed input error	Learned (dB)	HS-Rician (dB)	Learned loss vs. matched (dB)
None	-14.043	-12.496	0.000
B amplitude -10%	-11.613	-9.799	2.430
B amplitude $+10\%$	-12.252	-11.328	1.791
B phase -10°	-11.685	-10.730	2.358
B phase $+10^\circ$	-11.572	-10.564	2.471
Phase ramp $0 \to -10^\circ$	-13.267	-12.029	0.776
Phase ramp $0 \to +10^\circ$	-13.204	-11.774	0.839
Noise variance -6 dB	-13.806	-12.567	0.238
Noise variance $+6$ dB	-10.995	-12.404	3.048

Constant reference-phase errors rotate every estimated channel by the same phase; replay verifies the equivariance and unchanged phase-aligned errors. Alignment uses truth only as a diagnostic and is not an available inference correction. Absolute-frame errors in the table remain the reported errors.

D. Depth and Controller Trajectories

The stage-five output is taken from the unchanged ten-stage forward pass, with its original stage feature $(t+1)/10$. It is not a five-stage model retrained with $(t+1)/5$. Completing ten stages improves the seven-SNR average by 0.890 dB, with interval $[0.815, 0.964]$ dB; it does not help every cell. This prefix experiment does not determine an optimal depth or answer whether a trained twenty-stage model is better.

At the ordinary SNR 5 dB condition the mean relative threshold falls from about 0.915 at stage one to 0.362 at stage ten. Mean mixing is about 0.99996. Thus the useful adaptation is primarily in threshold strength, not abstention. Large relative corrections in a failure condition do not prove which feature or prior caused the failure.

V. RELATED STRUCTURAL AND SPARSE METHODS

OptShrink estimates singular-value weights from the observed noise spectrum [2]. LRHD combines that principle with repeated Hankel projection [3]; our one-pass post-processing diagnostic is not a reproduction of the full LRHD algorithm. In both cases the starting problem is noisy matrix denoising, not recovery from affine magnitudes. The noise on a Hankel matrix formed from an iterated channel estimate need not meet asymptotic independent-entry assumptions. Neither those guarantees nor universal optimality are asserted here.

Structured alternating least squares reconstructs low-rank matrices from linear measurements, including Hankel structure [4]. HSGD addresses partially observed Hankel matrices with sparse gross corruptions [5]. Their losses, observations and guarantees cannot be transferred unchanged to our nonlinear Rician model. They remain important alternative constructions, not irrelevant methods.

Noncoherent compressive channel estimation combines phase retrieval and sparse reconstruction using a designed sensing cascade [6]. GP-bandit beam training uses adaptive near-field beams, Rician-debiased pseudo-amplitudes and sparse phase retrieval [7]; it has no known additive reference.

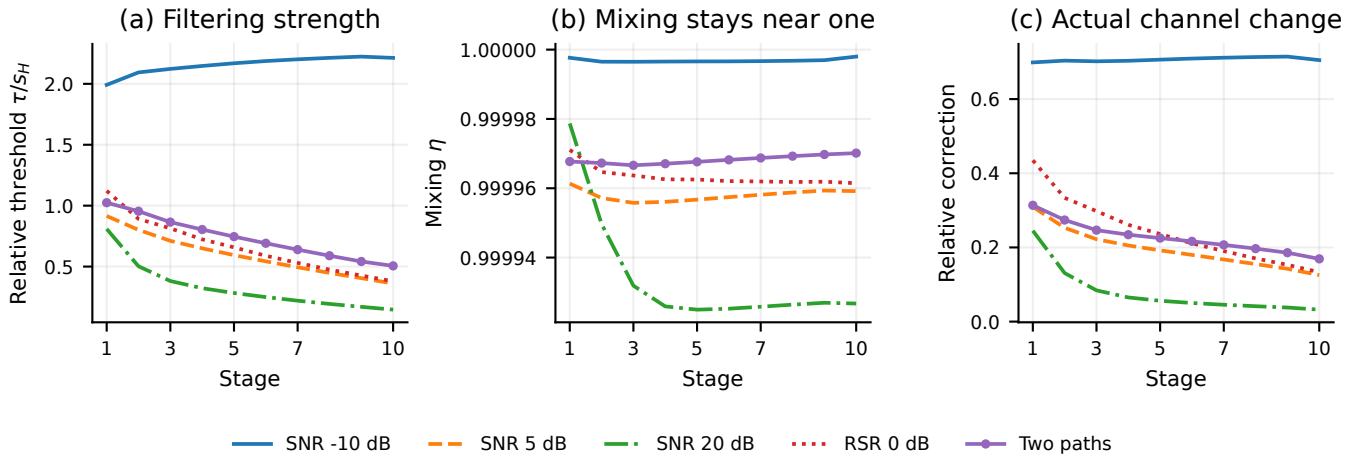


Fig. 2. New frozen-model trajectories, averaged over 32 cases, three model seeds and users. Main SNR curves use RSR 12 dB; weak-reference and two-path curves use SNR 5 dB. All other settings are unchanged. Mixing stays near one, while threshold and actual correction vary. Curves describe outputs; they do not establish causal feature importance or a failure detector.

Partially coherent matching pursuit retains complex observations with a shared unknown phase within each packet [8]. That information is absent from our magnitude-only inputs. A fair benchmark must adapt these methods to the same observation and pilot budget rather than importing their published scores.

The multi-shot Rydberg study [9] concerns signal-presence detection with Rician shots. It motivates accounting for observation statistics and shot budgets, but supplies neither channel-estimation results nor a heavy-tailed noise model for this experiment.

VI. What Is Still Not Tested

Depth. Ten stages were chosen as a fixed budget. A prefix of a ten-stage model is not a separately trained five-stage estimator. Running twenty steps would require a rule for new step sizes and stage features, or retraining. No trained 5/10/20 sensitivity study or depth optimum is claimed. A model with the same shared controller and T learned steps would have $178 + T$ parameters.

Noise and hardware. The original model has independent complex Gaussian noise before magnitude extraction. Incorrect variance inputs do not test non-Gaussian noise. Finite shot count alone does not imply heavy-tailed noise, and no cited detection model establishes that assumption for this receiver. Optical readout, atomic dynamics and hardware data remain untested. A Rician loss is not automatically matched to an alternative optical or multi-shot measurement statistic.

Structure. Cubic shrinkage is one design choice, not a proved optimal denoiser. Full LRHD, OptShrink-driven iterative reconstruction, structured ALS and robust Hankel completion require properly adapted measurement models and independently selected settings. Existing hard, global, adaptive and no-filtering controls do not exhaust these choices. An angle-domain sparse phase-retrieval method is also a relevant missing comparison, particularly for two-path channels. No superiority over those methods, or over published URformer, is claimed.

Training. Two-path and RSR 5 dB cases already occur in training; RSR 0 dB does not. Reweighting or curriculum training could alter the trade-off but has not been evaluated. No safety-critical protection guarantee follows from the average gains. The original pointwise intervals condition on fixed models and shared training data; the fresh-data follow-up is limited and does not estimate broad training-population uncertainty.

Array geometry. A two-dimensional array may suggest multilevel or block-Hankel structure, but its lifting dimensions, normalization, computational cost and training transfer must be defined anew. No 2D array result is reported.

VII. Initialization and Numerical Record

For array element n, form $C_n \in \mathbb{C}^{P\times(K+1)}$ with rows $[s_p^T, b_{np}]$. Let u_n be the dominant eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$ and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{13}$$

The first K components of $c_n u_n \exp[-j \arg(u_{n,K+1})]$ initialize the channel row. Only the 21 input pilots enter this calculation during training. The withheld columns do not enter the initial estimate.

The original 43 focused tests cover likelihood and derivative checks, noise scaling, finite gradients, repeated singular values, pilot-split isolation, single/double precision agreement and deterministic recovery. The submission audit recalculated 1,120 main, 702 follow-up and 432 input-study result files. Three primary checkpoints replayed eight saved observations exactly; 24 input-study checkpoint replays were also checked. This is not a replay of every historical experiment or an independent hardware validation.

One finite-input SVD failure occurred in the original classical HS-GS test at $N = 64$. The same matrix was recomputed

with SciPy's alternative SVD driver; relative reconstruction error was below 3×10^{-15}. The trial was retained with unchanged inputs and iteration budget. Original software versions were Python 3.12, NumPy 2.4.6, SciPy 1.17.1 and PyTorch 2.13.0. The original compact reproducibility package does not include every full observation/prediction array.

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